

Martin Fleischmann

¹Urban and Regional Laboratory, Charles University, CZ

²Geographic Data Science Lab, University of Liverpool, UK

ORCID: 0000-0003-3319-3366

Researcher ID: HKN-7973-2023

EDUCATION

Ph.D. Architecture, University of Strathclyde, Glasgow, 2021

MSc. Urban Design, University of Strathclyde, Glasgow, 2017

ACADEMIC APPOINTMENTS

2026– Assistant Professor, Dept. of Social Geography and Regional Development, Charles University

2024–26 Research Associate, Dept. of Social Geography and Regional Development, Charles University

2023–24 Postdoctoral Research Associate, The Alan Turing Institute, UK

2022–23 Researcher, Dept. of Social Geography and Regional Development, Charles University

2020–22 Research Associate, Dept. of Geography and Planning, University of Liverpool

HONORARY APPOINTMENTS

2022– Fellow, Dept. of Geography and Planning, University of Liverpool

RESEARCH INCOME

2026 *Software Sustainability Institute* ‘Geospatial Maintenance for Urban Social and Environmental Sciences (GeoMUSES)’, £150,000

2024 *European Space Agency* ‘EuroFab: European Urban Fabric Classification Using Artificial Intelligence’, €250,000

2023 *Primus Research Programme, Charles University* ‘Influence of Socioeconomic and Cultural Factors on Urban Structure in Central Europe’, €325,000

PUBLICATION ACTIVITY

24 papers indexed in Web of Science, 2 papers in other peer-reviewed journals, 5 policy reports

Web of Science: 728 citations, H-index: 11, Google Scholar: 2147 citations, H-index: 14

Selected publications not included in Bibliography

2026 Metzler, B., Fleischmann, M. and Arribas-Bel, D., ‘Is a (satellite) image worth a thousand data points? Comparing machine learning approaches to predict environmental and social inequalities in England’ *PLoS One* 21, 8. doi:10.1371/journal.pone.0356472

2026 Fleischmann, M., Van Den Bossche, J., Jordahl, K., et al., ‘GeoPandas: Fundamental Data Structures for Vector Spatial Data in Python’ *Computers, Environment and Urban Systems* 130, 102495. doi:10.1016/j.compenvurbsys.2026.102495

2026 Majer, H., and Fleischmann, M. ‘Predicting local climate zones using urban morphometrics and satellite imagery’ *Remote Sensing Applications: Society and Environment* 43, 102135 doi:10.1016/j.rsase.2026.102135

2026 Fleischmann, M., Samardzhiev, K., Brázdová, A., Dančejová, D., and Winkler, L., ‘The hierarchical morphotope classification: A theory-driven framework for large-scale analysis of built form’ *Cities* 174, 107047 doi:10.1016/j.cities.2026.107047

2026 Fleischmann, M., Vybornova, A., Gaboardi J., Brázdová, A., and Dančejová, D., ‘Adaptive continuity-preserving simplification of street networks’ *Computers, Environment and Urban Systems* 123, 102354 doi:10.1016/j.compenvurbsys.2025.102354

- 2025 Pebesma, E., Fleischmann, M., Parry, J., Nowosad, J., Graser, A., Dunnington, D., Pronk, M., Schouten, R., Lovelace, R., Appel, M., and Abad, L., 'Spatial Data Science Languages: Commonalities and Needs' *Journal of Spatial Information Science* 31, 119–144. doi:10.5311/JOSIS.2025.31.462
- 2024 Fleischmann, M. and Arribas-Bel, D., 'Decoding (urban) form and function using spatially explicit deep learning' *Computers, Environment and Urban Systems* 112, 102147 doi:10.1016/j.compenvurbsys.2024.102147
- 2024 Fleischmann, M., and Vybornova, A., 'A shape-based heuristic for the detection of urban block artifacts in street networks' *Journal of Spatial Information Science* 28. doi:10.5311/JOSIS.2024.28.319
- 2023 Fleischmann, M. 'Clustergram: Visualization and diagnostics for cluster analysis'. *Journal of Open Source Software* 8 (89), 5240. doi:10.21105/joss.05240
- 2023 Wang, J., Fleischmann, M., Venerandi, A., Romice, O., Kuffer, M., and Porta, S. 'EO + Morphometrics: Understanding cities through urban morphology at large scale' *Landscape and Urban Planning*, 233. doi:10.1016/j.landurbplan.2023.104691
- 2022 Rowe, F., Calafiore, A., Arribas-Bel, D., Samardzhiev, K., and Fleischmann, M. 'Urban exodus? Understanding human mobility in Britain during the COVID-19 pandemic using Meta-Facebook data' *Population, Space and Place*, e2637. doi:10.1002/psp.2637
- 2022 Arribas-Bel, D. and Fleischmann, M. 'Spatial Signatures: Understanding (urban) spaces through form and function' *Habitat International*, 128, (102641). doi:10.1016/j.habitatint.2022.102641
- 2022 Fleischmann, M. and Arribas-Bel, D. 'Geographical characterisation of British urban form and function using the spatial signatures framework' *Scientific Data*, 9, (546). doi:10.1038/s41597-022-01640-8
- 2022 Venerandi, A., Feliciotti, A., Fleischmann, M., Kourtiti, K., Porta, S. 'Urban form character and Airbnb in Amsterdam *N L*: A morphometric approach' *Environment and Planning B: Urban Analytics and City Science* 50 (2), 386-400. doi:10.1177/23998083221115196
- 2020 Fleischmann, M., Romice, O. and Porta, S. 'Measuring urban form: overcoming terminological inconsistencies for a quantitative and comprehensive morphologic analysis of cities.' *Environment and Planning B: Urban Analytics and City Science* 48 (8), 2133-2150. doi:10.1177/2399808320910444

RESEARCH SOFTWARE DEVELOPMENT

- 2026– SpML: Spatial modeling based on scikit-learn. (author and maintainer)
- 2022– Xvec: vector data cubes for Xarray. (author and maintainer)
- 2020– clustergram: visualization and diagnostics for cluster analysis (author and maintainer)
- 2020– PySAL: Python Spatial Analysis Library (core developer)
- 2019– GeoPandas: an open source project to make working with geospatial data in python easier (core developer)
- 2018– momepy: Urban Morphology Measuring Toolkit (author and maintainer)

EDITORIAL APPOINTMENTS

- 2025–26 Computers, Environment and Urban Systems
Guest Editor of a Special Issue on Open Urban Data Science
- 2021–25 Journal of Open Source Software
Associate Editor
- 2026– Environment and Planning B: Urban Analytics and City Science; European Journal of Geography
Member of Editorial Board

HONORS AND AWARDS

- 2024 Dean's Award for Early Career Researcher, Faculty of Science, Charles University
- 2023 Michael Breheny Prize, Environment and Planning B: Urban Analytics and City Science

Updated October 2026